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1. Let $A = \{1, 2, \{3, 4\}, \{5\}\}$. Decide which of each of the following statements are true or false.

- (a) $1 \in A$
- (b) $\{1\} \in A$
- (c) $\{5\} \in A$
- (d) $\{1, 2\} \in A$
- (e) $\{1, 2\} \subseteq A$
- (f) $\emptyset \in A$

Solution: Enter your solution here:

(a) $1 \in A$ **True.** The element 1 is listed explicitly as a member of A .

(b) $\{1\} \in A$ **False.** The set $\{1\}$ is not one of the four elements listed in A ; only the element 1 itself belongs to A .

(c) $\{5\} \in A$ **True.** The set $\{5\}$ is listed explicitly as a member of A .

(d) $\{1, 2\} \in A$ **False.** The set $\{1, 2\}$, taken as a single object, is not one of the four elements listed in A .

(e) $\{1, 2\} \subseteq A$ **True.** Both $1 \in A$ and $2 \in A$, so every element of $\{1, 2\}$ is an element of A .

(f) $\emptyset \in A$ **False.** The empty set is not one of the four elements listed in A .

2. Consider the following logical equivalence:

Theorem. For all logical statements P , Q , and R , we have

$$(P \Rightarrow Q) \Rightarrow R \equiv ((\neg P) \Rightarrow R) \wedge (Q \Rightarrow R).$$

Prove Theorem 1 using previously established logical equivalences.

Solution: Enter your solution here:

Proof. We assume that P , Q , and R are logical statements, and we will show that $(P \Rightarrow Q) \Rightarrow R$ is logically equivalent to $((\neg P) \Rightarrow R) \wedge (Q \Rightarrow R)$ by simplifying the right-hand side with a chain of logical equivalences until it matches the left-hand side.

$((\neg P) \Rightarrow R) \wedge (Q \Rightarrow R) \equiv (\neg \neg P \vee R) \wedge (\neg Q \vee R)$	Implication (applied twice)
$\equiv (P \vee R) \wedge (\neg Q \vee R)$	Double Negation
$\equiv (P \wedge \neg Q) \vee R$	Distributive
$\equiv (\neg \neg P \wedge \neg Q) \vee R$	Double Negation
$\equiv \neg(\neg P \vee Q) \vee R$	De Morgan
$\equiv \neg(P \Rightarrow Q) \vee R$	Implication
$\equiv (P \Rightarrow Q) \Rightarrow R$	Implication

Since each step above is a logical equivalence, we conclude that

$$(P \Rightarrow Q) \Rightarrow R \equiv ((\neg P) \Rightarrow R) \wedge (Q \Rightarrow R).$$

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3. Use set builder notation to write down the following sets:

- (a) The set A of all odd positive integers.
- (b) The set B of all real numbers strictly larger than π .
- (c) The set C given by

$$C = \left\{ \dots, 2^{-4}, 2^{-3}, 2^{-2}, \frac{1}{2}, 1 \right\}.$$

Solution: Enter your solution here:

(a)

$$A = \{x \mid x = 2k + 1, k \in \mathbb{Z}, k \geq 0\}$$

(b)

$$B = \{x \in \mathbb{R} \mid x > \pi\}$$

(c)

$$C = \{x \mid x = 2^n, n \in \mathbb{Z}, n \leq 0\}$$

4. Consider the following statements:

- (a) There exists a rational number that is less than or equal to -1010 .
- (b) For all integers x and y , we have $xy > x$.
- (c) For all real numbers ϵ , there exists a real number δ such that if $|x - \pi| < \delta$, then $|\sin(x)| < \epsilon$.

For each statement, complete the following three tasks:

- Rewrite the statement in symbolic form.
- Write its negation in symbolic form.
- Write a useful negation of each of the following statements as English sentences, without symbols for quantifiers, but where you can use symbols as they were used in the original statements.

Solution: Enter your solution here:

(a) There exists a rational number that is less than or equal to -1010 .

Symbolic form:

$$\exists x \in \mathbb{Q} (x \leq -1010)$$

Negation (symbolic form):

$$\neg[\exists x \in \mathbb{Q} (x \leq -1010)] \equiv \forall x \in \mathbb{Q} (x > -1010)$$

Useful negation (English sentence): Every rational number x satisfies $x > -1010$.

(b) For all integers x and y , we have $xy > x$.

Symbolic form:

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z} (xy > x)$$

Negation (symbolic form):

$$\neg[\forall x \in \mathbb{Z} \forall y \in \mathbb{Z} (xy > x)] \equiv \exists x \in \mathbb{Z} \exists y \in \mathbb{Z} (xy \leq x)$$

Useful negation (English sentence): There exist integers x and y such that $xy \leq x$.

(c) For all real numbers ϵ , there exists a real number δ such that if $|x - \pi| < \delta$, then $|\sin(x)| < \epsilon$.

Symbolic form:

$$\forall \epsilon \in \mathbb{R} \exists \delta \in \mathbb{R} (|x - \pi| < \delta \rightarrow |\sin(x)| < \epsilon)$$

Negation (symbolic form):

$$\neg[\forall \epsilon \in \mathbb{R} \exists \delta \in \mathbb{R} (|x - \pi| < \delta \rightarrow |\sin(x)| < \epsilon)] \equiv \exists \epsilon \in \mathbb{R} \forall \delta \in \mathbb{R} (|x - \pi| < \delta \wedge |\sin(x)| \geq \epsilon)$$

Useful negation (English sentence): There exists a real number ϵ such that for every real number δ , we have $|x - \pi| < \delta$ and $|\sin(x)| \geq \epsilon$.